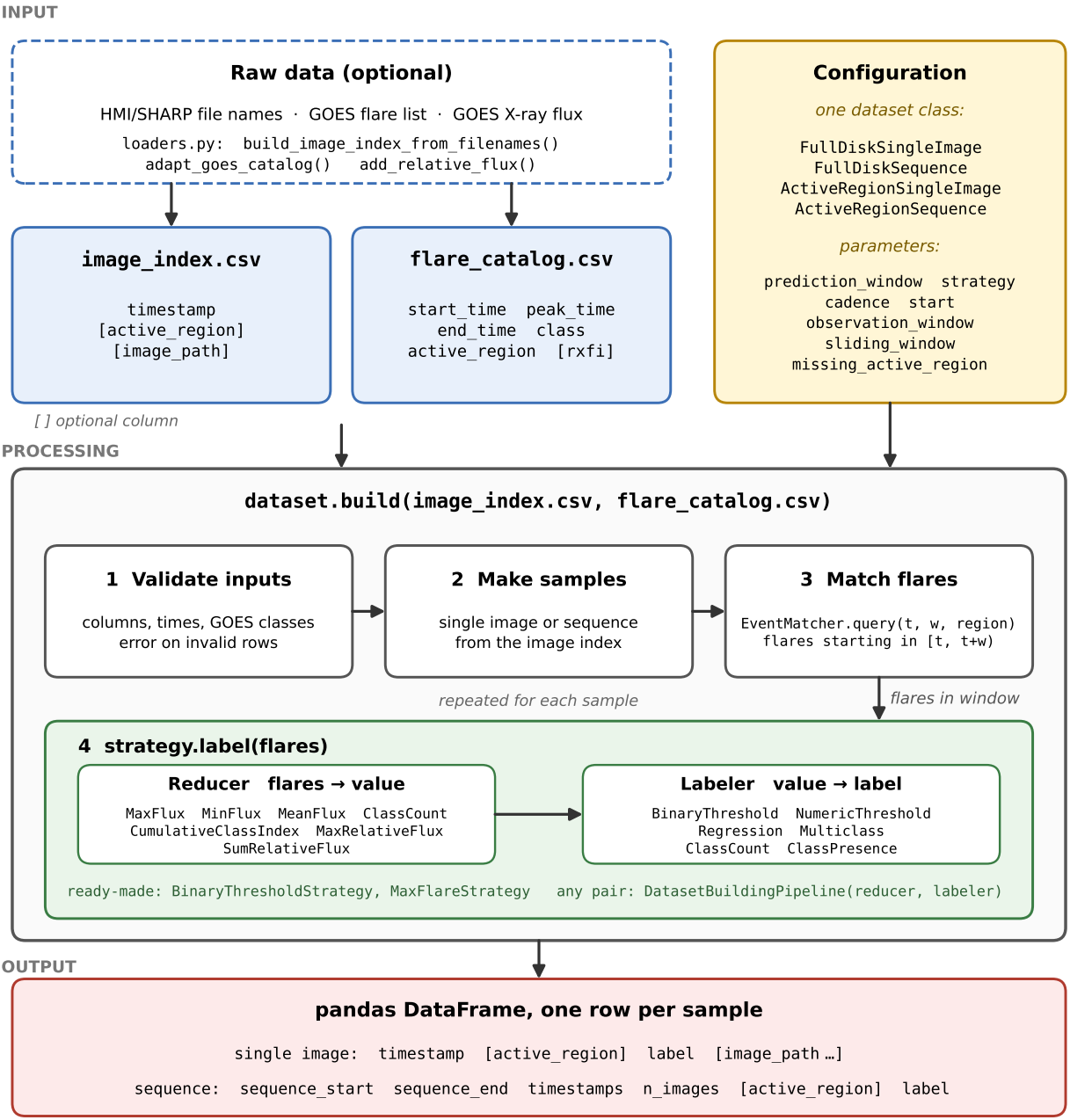

INPUT
Raw data (optional)
HMI/SHARP file names · GOES flare list · GOES X-ray flux
loaders.py: build_image_index_from_filenames()
adapt_goes_catalog() add_relative_flux()
image_index.csv
timestamp
[active_region]
[image_path]
flare_catalog.csv
start_time peak_time
end_time class
active_region [rxfi]
Configuration
one dataset class:
FullDiskSingleImage
FullDiskSequence
ActiveRegionSingleImage
ActiveRegionSequence
parameters:
prediction_window strategy
cadence start
observation_window
sliding_window
missing_active_region
[ ] optional column
PROCESSING
dataset.build(image_index.csv, flare_catalog.csv)
1 Validate inputs
columns, times, GOES classes
error on invalid rows
2 Make samples
single image or sequence
from the image index
3 Match flares
EventMatcher.query(t, w, region)
flares starting in [t, t+w)
repeated for each sample
flares in window
4 strategy.label(flares)
Reducer flares → value
MaxFlux MinFlux MeanFlux ClassCount
CumulativeClassIndex MaxRelativeFlux
SumRelativeFlux
Labeler value → label
BinaryThreshold NumericThreshold
Regression Multiclass
ClassCount ClassPresence
ready-made: BinaryThresholdStrategy, MaxFlareStrategy
any pair: DatasetBuildingPipeline(reducer, labeler)
OUTPUT
pandas DataFrame, one row per sample
single image: timestamp [active_region] label [image_path …]
sequence: sequence_start sequence_end timestamps n_images [active_region] label